The Product Rule: The Product Rule states that:

$$\frac{d}{dx}\left[u(x)v(x)\right] = _____$$

Antidifferentiating both sides, we get

$$u(x)v(x) = \int _____ + \int _____.$$

We can now write $du = u'(x)\,dx$ and $dv = v'(x)\,dx$:

$$u(x)v(x) = \int _____ + \int _____.$$

Rearranging the terms and dropping the x's, we get the

Integration by Parts Formula:

$$_____$$

In order for this to be useful, the integral on the right needs to be easier than the integral on the left. How do we pick u and dv? Well, to start with, dv should to be easy to integrate!

Examples

$\int xe^x\,dx$

$u = ____$ $dv = _____$
$du = _____$ $v = ____$

$\int_1^2 \ln x\,dx$

$u = _____$ $dv = _____$
$du = _____$ $v = ____$

Questions

Most (but not all!) of these are integration by parts problems. For some, you will need to integrate by parts more than once. For each *indefinite* integral, check the solution by differentiating. Integrals done using by-parts usually require the product rule to check them.

1. $\int x \ln x \, dx$

□Regrade

2. $\int_0^{\pi} x \sin x \, dx$

□Regrade

3. $\int_0^1 x\sqrt{x+1}\,dx$ (Compare to the same question on worksheet 7-2. Which method do you prefer?) □Regrade

4. $\int_1^2 x^2 e^x \, dx$ (Hint: Integrate by parts twice. Exact answer only please.)

□Regrade

5. $\int \arctan x \, dx$ (Hint: Let $u = \arctan x$. You'll need to use both integration by parts and a substitution here to complete all the work.)

□Regrade

6. $\int e^x \cos x \, dx$ (Hint: Integrate by parts twice, and carefully observe the equation you obtain.)

□Regrade

7. $\int x(\ln x)^3\,dx$ (Hint: Integrate by parts three times. I suggest carefully separating out your work to keep track of it. When checking your answer, do each term separately.)
□Regrade

8. $\int xe^{-x^2}\,dx$ (Hint: Don't work too hard!)

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